

Synaptic Plasticity as Short-Term Memory

One-page concept summary · Anand Tudu · DataForge 2026, Pathway track · Companion artifact: bdh-synaptic-memory-explainer.vercel.app

Central claim (falsifiable). Attention can be reformulated as synaptic memory — recent token interactions temporarily strengthen local connection weights through Hebbian updates — letting a fixed-size recurrent state carry a sequence model’s working memory without a KV-cache that grows linearly with context length. The companion artifact reproduces the claim in a 24×24 simulation: if overlap-driven recall errors did not grow with sparse co-writes, or if memory use rose with context length, the claim would fail.

The design pressure, and the one-line mechanism (formal + classical theory)

Every Transformer pays an inference-time scaling tax: the KV-cache stores each token’s keys and values, so memory grows linearly with context while attention compute grows quadratically — the first wall long documents and multi-hour agents hit. The pivotal formal result is Schlag, Irie & Schmidhuber (2021): linear attention’s recurrent state is a **fast-weight program**, where each token performs a rank-1 outer-product write and attention is the read-back. Caching and plasticity are therefore the same computation under different policies — a KV-cache is $\lambda = 1$ with verbatim writes; a synaptic state is $\lambda < 1$ with sparse writes.

The mechanism is one line: sparse, non-negative activations $\mathbf{x}$ (in reported BDH runs roughly 5% of neurons fire, varying with predictability) write $\mathbf{W} \leftarrow \lambda \cdot \mathbf{W} + \eta \cdot (\mathbf{x} \cdot \mathbf{x}^m)$ into a fixed matrix; recall is content-addressed, $\mathbf{y} = \mathbf{W} \cdot \text{cue}$; multiplicative decay makes the state short-term. Three costs follow: **interference** (overlapping patterns corrupt each other’s recall), **bounded capacity** ($\approx 0.14 \cdot d$ patterns for outer-product rules; Hopfield, 1982), and **weak order information** — outer-product traces barely distinguish “dog bites man” from “man bites dog.”

Where the concept lives in BDH and BDH-CQ (developer-reported)

In **BDH** (Kosowski et al., 2025) the mechanism is the memory substrate: working memory lives in a fixed synaptic state on a scale-free graph of spiking neurons, attention is reformulated as synaptic memory through Hebbian writes, and the GPU variant **BDH-GPU** is a ReLU low-rank formulation with linear attention that follows Transformer scaling laws. Reported results: BDH rivals GPT-2 on language and translation at 10M–1B parameters on the same training data, and synapses behave monosemantically — individual connections strengthen for specific concepts, making state inspectable by construction. No public checkpoint or independent reproduction exists yet.

BDH-CQ (Engdahl et al., 2026) extends state-as-memory from session memory to *task adaptation*: inference-time inputs continuously update recurrent memory, and the answer is produced by iterative computation in latent space — no verbalized chain of thought, no parameter updates at inference. Its contextual memory has an additive-per-demonstration accumulation rule, exactly the special case the fast-weight/linear-attention view predicts. Reported result: a 150M-parameter configuration reaches 29.5% pass@2 on the public ARC-AGI-1 set at a computed \$0.0007 per task, which its authors state breaks the previously reported cost-accuracy Pareto frontier. This is a developer-reported benchmark result, not an external evaluation — but it is direct evidence that a plastic fixed state can carry adaptation, not merely session memory.

Competitive context

System	Working memory	Write rule	Verbatim recall	Order info	Evidence level
Transformer + KV-cache	Grows O(T)	Append keys/values	Yes	Yes (positions)	standard practice
Mamba (selective SSM)	Fixed size	Gated selective state update	No	Yes	dev-reported, up to 8-scale
LRU / gated linear RNN	Fixed size	Gated linear recurrence	No	Yes	peer-reviewed, ICML 2024
BDH (synaptic)	Fixed synapse matrix	Sparse Hebbian write × decay	No	Weak (naive form)	dev-reported, 10M–1B

Strengths, losses, and the open question

Wins: constant per-sequence memory (the same state serves token 100 and token 100,000); interpretability by construction — the synapse matrix is the memory, so state is inspected rather than post-hoc explained; and one computational fabric serving memory, reasoning, and adaptation (BDH-CQ). **Losses:** verbatim recall, order-sensitive reasoning, and exact history — caching wins these outright, which is why pragmatic stacks are hybrid: cache what must be exact, let the rest decay through plastic state. **The open question is capacity scaling:** whether plastic states can hold long, order-dependent context at frontier scale. BDH’s pretraining-scaling experiments beyond 1B parameters are early and developer-reported, and none of the results above has been independently reproduced — treating them as settled would overclaim.

Primary sources & where to continue

- Kosowski et al. (2025). *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*. arXiv:2509.26507 — introduces BDH; synaptic memory substrate, scaling results. Code: github.com/pathwaycom/bdh.
- Engdahl et al. (2026). *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888 — state-as-memory for task adaptation; ARC-AGI-1 result.
- Schlag, Irie & Schmidhuber (2021). *Linear Transformers Are Secretly Fast Weight Programmers*. ICML. arXiv:2102.11174 — the formal caching↔plasticity equivalence.
- Gu & Dao (2023). *Mamba: Linear-Time Sequence Modeling with Selective State Spaces*. arXiv:2312.00752 — fixed-state alternative at scale.
- Orvieto et al. (2024). *Resurrecting RNNs for Long Sequences*. ICML. arXiv:2303.06349 — which ingredients (gating, decay, normalization) close the gap.
- Hopfield (1982). *PNAS* 79(8). doi:10.1073/pnas.79.8.2554 — outer-product capacity bound.

Prepared with AI assistance (Codebuff) and reviewed by the author; all claims verified against the cited primary sources and labeled by evidence level: *formal* (proven equivalence), *peer-reviewed*, *developer-reported* (no independent reproduction). The companion artifact is a labeled toy of the mechanism, not an official BDH model.